

OFTEN TIMES THE INPUT SIGNAL IS RATHER LONG, ESPECIALLY IN REAL-TIME DSPS.

WE NEED TO FIND A COMPUTATIONALLY EFFICIENT METHOD TO PERFORM A LINEAR CONVOLUTION.

THE STRATEGY CONSISTS OF SEGMENTING THE SIGNAL INTO BLOCKS OF FIXED LENGTH AND COMPUTING THE ALGORITHM FOR EACH BLOCK SEPARATELY.

THERE ARE MAINLY TWO APPROACHES:

- OVERLAP-ADD
- OVERLAP-SAVE

OVERLAP-ADD

IT EXPLOITS THE PROPERTY OF LINEARITY OF THE CONVOLUTION

$$Y[n] = h[n] * [x_1[n] + x_2[n]] = h[n] * x_1[n] + h[n] * x_2[n]$$

BY DIVIDING THE INPUT SIGNAL $x[n]$ INTO MULTIPLE NON OVERLAPPING BLOCKS $x_m[n]$ OF LENGTH L AND THEN CONVOLVING EACH BLOCK WITH $h[n]$ TO PRODUCE $Y_m[n]$.

$$L=4 \quad x[n] = \sum_{m=1}^4 x_m[n]$$

$$x[n] = \begin{matrix} n=0 \\ \downarrow \\ [2 \ 3 \ 4 \ 7 \ 9 \ 3 \ 7 \ 6 \ 8 \ 7 \ 2 \ 4 \ 4 \ 1 \ 1 \ 1] \end{matrix}$$

$$x_1[n] = [2 \ 3 \ 4 \ 7 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$x_2[n] = [0 \ 0 \ 0 \ 0 \ 9 \ 3 \ 7 \ 6 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$x_3[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 8 \ 7 \ 2 \ 4 \ 0 \ 0 \ 0 \ 0]$$

$$x_4[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 4 \ 1 \ 1 \ 1]$$

$$h[n] * x[n] = h[n] * x_1[n] + h[n] * x_2[n] + h[n] * x_3[n] + h[n] * x_4[n]$$

$$h[n] = [0 \ 1 \ 0 \ 0 \ 0] = \delta[n-1]$$

$$N = 4 + 2 - 1 = 5$$

$$x[n] = \begin{matrix} n=0 \\ \downarrow \\ [2 \ 3 \ 4 \ 7 \ 9 \ 3 \ 7 \ 6 \ 8 \ 7 \ 2 \ 4 \ 4 \ 1 \ 1 \ 1] \end{matrix}$$

$$x_1[n] = [2 \ 3 \ 4 \ 7 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$x_2[n] = [0 \ 0 \ 0 \ 0 \ 9 \ 3 \ 7 \ 6 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$x_3[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 8 \ 7 \ 2 \ 4 \ 0 \ 0 \ 0 \ 0]$$

$$x_4[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 4 \ 1 \ 1 \ 1]$$

$$Y_1[n] = [0 \ 2 \ 3 \ 4 \ 7 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$Y_2[n] = [0 \ 0 \ 0 \ 0 \ 9 \ 3 \ 7 \ 6 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$Y_3[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 8 \ 7 \ 2 \ 4 \ 0 \ 0 \ 0 \ 0]$$

$$Y_q[n] = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 4 \ 1 \ 1 \ 1]$$

$$Y[n] = [0 \ 2 \ 3 \ 4 \ 1 \ 9 \ 3 \ 7 \ 6 \ 8 \ 7 \ 2 \ 4 \ 4 \ 1 \ 1 \ 1]$$

SINCE THE BLOCKS HAVE BEEN ZERO-PADDED, THE LINEAR CONVOLUTION IS EQUIVALENT TO THE CIRCULAR CONVOLUTION, MEANING THE ALGORITHM CAN BE RUN USING THE DFT AND MULTIPLYING IN THE FREQUENCY DOMAIN, WHICH IS MUCH MORE EFFICIENT

OVERLAP SAVE

IN CONTRAST TO THE OVERLAP-ADD METHOD WHICH SEPARATES A LONG SEQUENCE IN NON-OVERLAPPING BLOCKS, THE OVERLAP-SAVE METHOD SEPARATES IT INTO OVERLAPPING BLOCKS.

EACH BLOCK HAS LENGTH $N=L+M-1$ AND CONTAINS THE LAST $M-1$ POINTS OF THE PREVIOUS BLOCK $x_{m-1}[n]$ ($m>1$).

SINCE THE FIRST BLOCK HAS NO PREVIOUS BLOCK, THE FIRST $M-1$ POINTS ARE ZEROS.

$$M=2$$

$$L=4$$

$$N=L+M-1=5$$

$$h[n] = [0 \ 1] \begin{matrix} n=0 \\ \downarrow \end{matrix} \begin{matrix} 0 & 0 & 0 \end{matrix}$$

$$x[n] = [2 \ 3 \ 4 \ 1 \ 9 \ 3 \ 7 \ 6 \ 8 \ 7 \ 2 \ 4 \ 4 \ 1 \ 1 \ 1]$$

$$x_1[n] = [0 \ 2 \ 3 \ 4 \ 1]$$

$$x_2[n] = [1 \ 9 \ 3 \ 7 \ 6]$$

$$x_3[n] = [6 \ 8 \ 7 \ 2 \ 4]$$

$$x_4[n] = [4 \ 4 \ 1 \ 1 \ 1]$$

$$x_5[n] = [1 \ 0 \ 0 \ 0 \ 0]$$

REMEMBER ALSO THIS!

$$y_1[n] = [1 \ 0 \ 2 \ 3 \ 4]$$

$$y_2[n] = [6 \ 1 \ 9 \ 3 \ 7]$$

$$y_3[n] = [4 \ 6 \ 8 \ 7 \ 2]$$

$$y_4[n] = [1 \ 4 \ 4 \ 1 \ 1]$$

$$y_5[n] = [0 \ 1 \ 0 \ 0 \ 0]$$

$$Y[n] = [0 \ 2 \ 3 \ 4 \ 1 \ 9 \ 3 \ 7 \ 6 \ 8 \ 7 \ 2 \ 4 \ 4 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0]$$

ADD STORE RESULT

SAVE STORE RPTG

FFT

COMPUTING THE DFT FOR LARGE VALUES OF N IS VERY EXPENSIVE (N COMPLEX MULTIPLICATIONS + $N-1$ COMPLEX ADDITIONS $x_k \rightarrow O(N^2)$).

USING THE FFT ALGORITHM WE CAN REDUCE $O(N \log N)$!

FFT ALGORITHMS EXPLOIT THE SYMMETRY AND PERIODICITY OF THE COMPLEX EXPONENTIAL AND RELY ON THE FACT THAT AN N -POINT DFT CAN BE COMPUTED BY TWO $N/2$ -POINT DFTS

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad W_N = e^{-j \frac{2\pi}{N}}$$

$$W_N^k = e^{-j \frac{2\pi}{N} k} = (-1) \left(-e^{-j \frac{2\pi}{N} k} \right) = (e^{-j\pi}) \left(-e^{-j \frac{2\pi}{N} k} \right) = \left(e^{-j \frac{2\pi}{N} \cdot \frac{N}{2}} \right) \left(-e^{-j \frac{2\pi}{N} k} \right) \\ = -e^{-j \frac{2\pi}{N} \left(\frac{N}{2} + k \right)} = -W_N^{k + \frac{N}{2}}$$

$$W_N^k = 1 \cdot e^{-j \frac{2\pi}{N} k} = e^{-j2\pi} e^{-j \frac{2\pi}{N} k} = e^{-j \frac{2\pi}{N} \cdot N} e^{-j \frac{2\pi}{N} k} = e^{-j \frac{2\pi}{N} (k+N)} = W_N^{k+N}$$

$$W_N^k = -W_N^{k + \frac{N}{2}}$$

$$W_N^k = W_N^{k+N}$$

LET'S ANALYZE THE COMPLEXITY: DFT IS $O(N^2)$. IF WE ASSUME HALF A LENGTH, WE OBTAIN $O(N^2/4)$. THEREFORE IF WE COULD SPLIT THE SEQUENCE IN TWO SEQUENCES OF HALF LENGTH WE WILL OBTAIN $O(2(N/2)^2) = O(N^2/2)$.

WE THEREFORE NEED TO FIND A WAY TO SPLIT THE SEQUENCE.

LET'S SEPARATE THE EVEN AND ODD SAMPLES (CAREFUL, NOT EVEN/ODD PART, IT'S JUST SAMPLES WITH EVEN/ODD INDICES)

$$x[n] = [0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7] \begin{cases} \text{EVEN INDICES} \\ e[n] = [x[0], x[2], x[4], x[6]] = [0 \ 2 \ 4 \ 6] \\ o[n] = [x[1], x[3], x[5], x[7]] = [1 \ 3 \ 5 \ 7] \\ \text{ODD INDICES} \end{cases}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} = \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_N^{2nk} + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_N^{(2n+1)k}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_N^{2nk} + W_N^k \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_N^{2nk}$$

$$\rightarrow W_N^{2nk} = e^{-j \frac{2\pi}{N} 2nk} = e^{-j \frac{2\pi}{N/2} nk} = W_{N/2}^{nk}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_{N/2}^{nk} + W_N^k \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_{N/2}^{nk}$$

$$= \underbrace{\sum_{n=0}^{\frac{N}{2}-1} e[n] W_N^{nk}}_{\frac{N}{2}\text{-DFT}} + W_N^k \underbrace{\sum_{n=0}^{\frac{N}{2}-1} o[n] W_N^{nk}}_{\frac{N}{2}\text{-DFT}}$$

$$X[k] = E[k] + W_N^k O[k], \quad 0 \leq k < \frac{N}{2}$$

Now need to find other half

$$X[k + \frac{N}{2}] = E[k + \frac{N}{2}] + W_N^{k + \frac{N}{2}} O[k + \frac{N}{2}]$$

$$= \sum_{n=0}^{\frac{N}{2}-1} e[n] W_N^{n(k + \frac{N}{2})} + W_N^{k + \frac{N}{2}} \sum_{n=0}^{\frac{N}{2}-1} o[n] W_N^{n(k + \frac{N}{2})}$$

$$\rightarrow W_N^{k + \frac{N}{2}} = W_N^k W_N^{\frac{N}{2}} = W_N^k e^{-j\frac{2\pi}{N} \frac{N}{2}} = -W_N^k$$

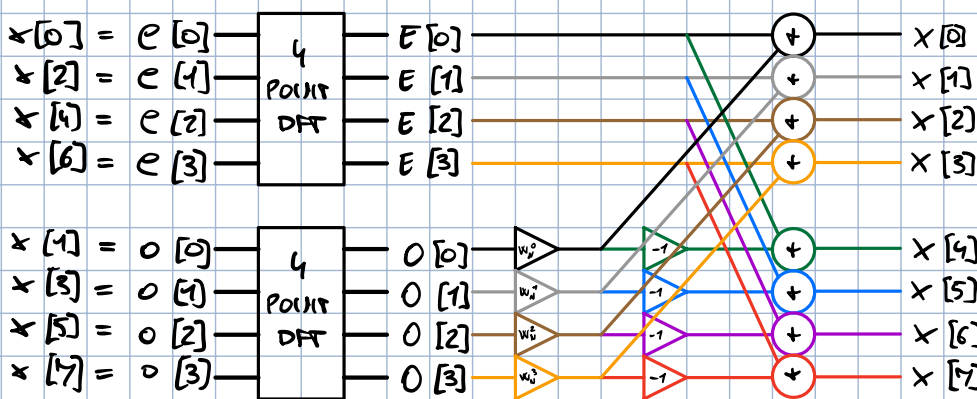
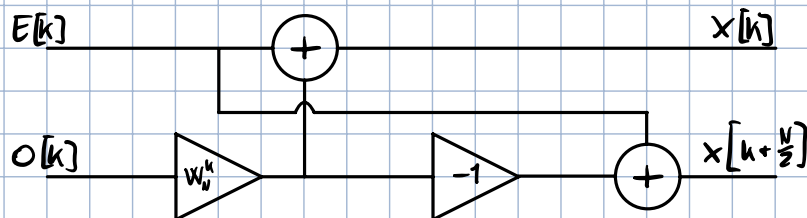
$$= \sum_{n=0}^{\frac{N}{2}-1} e[n] W_N^{n(k + \frac{N}{2})} - W_N^k \sum_{n=0}^{\frac{N}{2}-1} o[n] W_N^{n(k + \frac{N}{2})}$$

$$\rightarrow W_N^{n(k + \frac{N}{2})} = W_N^{nk} W_N^{n\frac{N}{2}} = W_N^{nk} \underbrace{e^{-j\frac{2\pi}{N} \frac{N}{2} n}}_{=1 \text{ if } n \text{ is even}} = W_N^{nk}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} e[n] W_N^{nk} - W_N^k \sum_{n=0}^{\frac{N}{2}-1} o[n] W_N^{nk}$$

$$X[k + \frac{N}{2}] = E[k] - W_N^k O[k], \quad 0 \leq k < \frac{N}{2}$$

BUTTERFLY DIAGRAM



$$X[k] = E[k] + W_8^k O[k]$$

$$X[k + \frac{N}{2}] = E[k] - W_8^k O[k]$$

THIS PROCESS CAN BE REITERATED RECURSIVELY TO OBTAIN THE 4-DFT USING TWO 2-DFTs.

$$E[n] = [x[0], x[2], x[4], x[6]]$$

$$E_e[n] = [x[0], x[4]]$$

$$E_o[n] = [x[2], x[6]]$$

$$E[k] = E_e[k] + W_4^k E_o[k], \quad 0 \leq k < \frac{N}{2}$$

$$E[k + \frac{N}{2}] = E_e[k] - W_4^k E_o[k], \quad 0 \leq k < \frac{N}{2}$$

$$O[k] = O_e[k] + W_4^k O_o[k]$$

$$O[k + \frac{N}{2}] = O_e[k] - W_4^k O_o[k]$$

WE CAN GO EVEN FURTHER FROM 2 TO 1

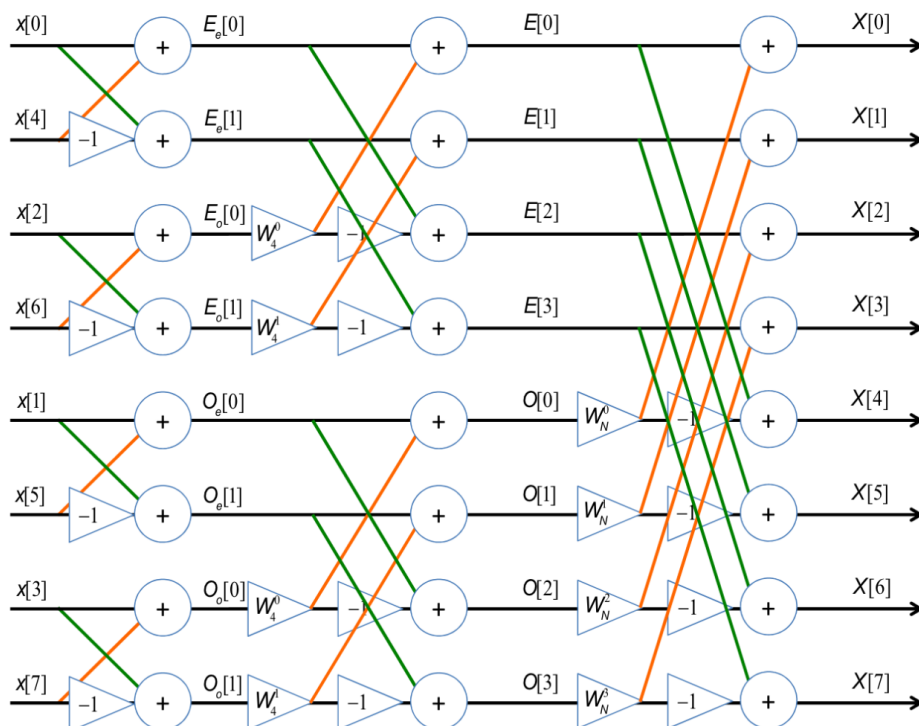
$$N=1 \quad X[0] = \sum_{n=0}^0 x[n] W_N^{0n} = x[0] \rightarrow 1\text{-POINT DFT IS THE SAMPLE ITSELF}$$

$$x[n] = [x[0], x[1]]$$

$$\begin{aligned} X[0] &= x[0] + x[1] \\ X[1] &= x[0] - x[1] \end{aligned} \rightarrow 2\text{-POINT DFT IS SIMPLE LINEAR COMBINATION}$$

RADIX-2

THE PROCESS OF DIVIDING N HALF EVERY TIME GIVES THE NAME "RADIX-2" TO THIS ALGORITHM.



STAGE 1

STAGE 2

STAGE 3

$\log_2 N$ STAGES